

# Junlin Wu

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## Education

**Columbia University** | M.S. in Data Science | Focus: Generative AI and LLM Applications    Sep 2023 - Dec 2024 | New York  
**University of California San Diego** | B.S. in Data Science, Economics Minor | GPA 3.9/4.0, Cum Laude    Sep 2019 - Mar 2023 | San Diego

## Professional Experience

**Huawei Technologies Co., Ltd.**    Aug 2025 - Present | Shanghai

**AI Engineer** | Computing Product Line, Lingqu Cluster Development Department

- Led quantization and accuracy tuning for **Flux, Wan 2.2, and Hunyuan** multimodal diffusion models on next-generation Ascend chips. Used operator profiling, activation distributions, and gradient signals to diagnose bottlenecks, integrate MindIE SD, and meet or exceed performance targets.
- Built a Pandas-based DiT-block profiling tool with fuzzy matching, cross-chip operator mapping, and configurable skip rules, cutting analysis time by about **50%**; adopted as a team standard across **5+ projects**.
- Owned end-to-end **Qwen3.5** vLLM adaptation for a new chip platform across bring-up, quantization, fused operators, prefill-decode disaggregation, deployment, and tuning; coordinated three engineering teams for on-time delivery and documented reusable serving practices.
- Built **SGLang** MXFP4/MXFP8 inference for diffusion, dense LLM, and MoE LLM workloads, including W4A4/W4A8 paths; supported online quantization and offline inference with msmodelslim pre-quantized weights.

**Columbia University Medical Center | Nectow Neuroscience Lab**    Nov 2023 - Dec 2024 | New York

**Data Analyst Intern**

- Built an **in vivo** electrophysiology pipeline for denoising, spike detection, and temporal alignment; modeled links between neural activity and feeding behavior to identify satiation-regulating neurons.
- Developed reusable Python analysis and visualization workflows; contributed data analysis and figures as a co-author to a study published in **Cell** (2025, DOI: 10.1016/j.cell.2025.01.018).

## Selected Projects

**TD Bank | Generative AI Executive Advisor**    TallMessiWu/td-llm-streamlit    Sep 2024 - Dec 2024 | New York

**Columbia University x TD Bank Capstone | Core Developer, 4-Person Team**

- Built a banking executive **RAG Agent** with LangChain, GPT-4, OpenAI Embeddings, Pinecone Serverless, and Streamlit, answering questions over reports and earnings-call transcripts with source citations.
- Created semantic rolling-window chunking with GPT-4 cleaning and SentenceTransformer topic boundaries; implemented **Top-100 semantic recall → filename-similarity filtering → Top-10 context** retrieval with file and page provenance.
- Added hallucination validation, evaluated relevance and groundedness with TruLens LLM-as-Judge, and traced the full retrieval-generation pipeline with **LangSmith**.

**From-Scratch ReAct Agent | GitHub Repository Analyzer**    TallMessiWu/react-agent    May 2026 | Shanghai

**Personal Project**

- Built a DeepSeek-powered **multi-step ReAct loop** without an Agent framework, using asyncio, httpx, and four Function Calling tools to explore GitHub repositories and generate structured reports.
- Added per-tool three-state circuit breakers, exponential backoff with **30% jitter**, and graceful degradation under rate limits, returning explainable structured fallbacks during API failures.
- Handled HTTP 429 responses, 301/302 redirects, and Base64 decoding; surfaced DeepSeek reasoning\_content to improve decision-chain observability.

## Technical Skills

**Advanced** Python (Pandas / NumPy), PyTorch, Agent / RAG (**ReAct, Function Calling, LangChain, Pinecone**), LLM inference and serving (**vLLM, SGLang**)

**Proficient** Model quantization (MXFP4 / MXFP8 / INT8), OpenAI / DeepSeek APIs, OpenAI Embeddings, TruLens, LangSmith, Streamlit, asyncio / httpx, GitHub REST API

**Languages** Mandarin Chinese (native), English (proficient)